COMMUNITY SERVICE PROJECT (CSP) | ACADEMIC YEAR 2024–2026 | ANDHRA PRADESH & TELANGANA

# Gramin Career Setu ([redacted])

A Bilingual, Offline-First Career Exploration, Vocational Counseling & ATS Resume Platform for Rural Youth

| | |
|---|---|
| **Project Category:** Community Service Project (CSP) | **Target Audience:** Rural Students, ITI / Polytechnic & Job Seekers |
| **Target Geography:** Rural AP & Telangana (Villages & Mandals) | **Supported Languages:** Telugu ([redacted]) & English (Dual-Mode) |
| **Architecture:** Offline-First PWA + Pure Node HTTP Server | **Quality Assurance:** 6/6 Automated Security & Functional Tests Passing |

**Executive Summary:** Gramin Career Setu is an end-to-end, zero-cost digital career empowerment platform engineered to overcome the informational, linguistic, and technological barriers that rural students face across Andhra Pradesh and Telangana. Mainstream job and counseling websites assume high-speed internet, modern laptops, and English fluency—leaving rural youth unaware of emerging sectors, vocational trades, government scholarships, and competitive examinations. Gramin Career Setu delivers an **offline-first, 100% vernacular-accessible web application** featuring real-time Telugu-English translation, a scientifically validated RIASEC career aptitude quiz, an algorithmic ATS resume builder with instant PDF generation, curated rural job listings, government welfare directories, educational roadmaps, and volunteer administration tools.

## 1. The Rural Career Crisis & Problem Statement

- **The Vernacular Divide:** Over 85% of rural secondary school students in AP and Telangana study in Telugu-medium institutions. Mainstream career platforms are exclusively in English, creating severe anxiety and excluding rural talent from discovering opportunities.
- **Bandwidth & Hardware Constraints:** Rural areas encounter power outages and spotty 2G/3G connectivity. Low-spec budget smartphones struggle with bloated 10MB+ JavaScript frameworks. A lightweight, offline-capable architecture is non-negotiable.
- **Absence of Professional Resumes:** Rural youth lack computer literacy and guidance to format resumes. Consequently, candidate applications are rejected outright by employer Applicant Tracking Systems (ATS).
- **Fragmented Awareness of Schemes:** Government welfare, fee reimbursements, and skill training programs (Jagananna Vidya Deevena, PMKVY, TS ePASS) are dispersed across complex official gazettes, preventing deserving candidates from claiming their entitlements.

## 2. Technical Specifications & Architectural Stack

| Layer | Technology / Component | Engineering Rationale & Architectural Role |
|---|---|---|
| **Frontend Runtime** | Pure HTML5 + Vanilla ECMAScript (ES6+) | Zero framework dependencies; loads under 120ms on low-spec Android phones. |
| **Design System** | CSS3 Custom Properties (tokens.css, saas.css) | Unified 8pt grid, WCAG 2.1 AA compliant contrast, dark/light theme support. |
| **Offline PWA** | Service Worker API + CacheStorage (sw.js) | Pre-caches all 21 views, fonts, images & scripts for 100% offline operation. |
| **i18n Engine** | Atomic Dictionary (i18n-dict.js, 248+ keys) | Zero-FOUC inline pre-paint; child-safe DOM node text replacement. |
| **PDF Generation** | Lazy-Loaded jsPDF (pdf-loader.js) | Dynamically loads 364 KB jsPDF only when user clicks export (0 KB initial cost). |
| **Local Server** | Hardened Node.js HTTP Server (server.js) | Protected against path traversal & malformed URIs; zero external npm deps. |
| **Test Automation** | Node.js Native Test Runner (node:test) | 6 automated specs checking security, 404s, pagination, and form limits. |

## 3. Core Functional Innovations & Solution Pillars

Gramin Career Setu is structured around 7 foundational pillars designed specifically to resolve grassroots rural employment and counseling hurdles:

### Pillar 1: Dual-Mode Bilingual i18n Translation Engine

A high-speed, client-side translation framework supporting real-time toggling between Telugu (■■■■■■) and English across all 21 pages. Features an atomic dictionary (assets/i18n-dict.js) with 248+ culturally vetted terms. Employs child-safe DOM text replacement that preserves icons, styles, and structural nodes while updating dynamic attributes (placeholder, aria-label, title). An inline pre-paint script checks localStorage on document initialization, completely eliminating Flash of Unstyled Content (FOUC).

### Pillar 2: Scientifically Grounded RIASEC Career Aptitude Quiz

A 12-question psychometric assessment adapted from John Holland's RIASEC framework (Realistic, Investigative, Artistic, Social, Enterprising, Conventional). Presents rural youth with practical situational scenarios (e.g., repairing machinery vs. organizing a village health camp vs. analyzing farm crop yields). The mathematical engine computes weighted dimensional scores, renders an interactive radar/spider chart, and highlights tailored career archetypes with concrete next steps.

### Pillar 3: Algorithmic ATS Resume Builder & Live A4 Canvas

A guided 7-step wizard: Personal Details, Education, Skills, Projects, Experience, Certifications, and Review. Provides live dynamic synchronization with a virtual A4 sheet, auto-saving drafts into localStorage. Computes a real-time ATS Compatibility Score (0–100%) checking keyword density and structured sections. Seamless client-side PDF export is powered by lazy-loaded jsPDF (assets/pdf-loader.js), ensuring zero network overhead until explicitly invoked.

### Pillar 4: Verified Rural & Regional Job Opportunities Board

Curated job catalog focusing on entry-level, semi-skilled, and technical openings in Agriculture, IT Services, Banking, Healthcare, and Logistics. Incorporates multi-criteria search, category filters, and pagination without full-page reloads. Displays transparent qualification prerequisites, wage ranges, and direct recruiter application channels.

### Pillar 5: Central & State Welfare Schemes & Scholarships Portal

Consolidates leading Andhra Pradesh and Telangana government youth empowerment programs (Jagananna Vidya Deevena, Vasathi Deevena, Ambedkar Overseas Vidya Nidhi, TS ePASS, PMKVY skill vouchers). Provides clean eligibility checklists, deadlines, and direct application links, demystifying bureaucratic procedures.

### Pillar 6: Visual Educational Roadmaps & Career Detail Catalog

Milestone-based educational trajectory maps guiding students from 10th Standard / Intermediate to competitive professions (Software Engineering, Civil Services, Banking, Police & Armed Forces, Modern Agriculture, and Nursing). Breaks down mandatory entrance tests (EAPCET, POLYCET, UPSC, APPSC), training duration, cost estimates, and salary ladders.

### Pillar 7: Community Field Management & Impact Analytics

A volunteer administration portal (admin/index.html) enabling village coordinators to manage registrations, track quiz completion statistics, and synchronize records with Google Sheets for central college reporting and offline survey backup.

## 4. Comprehensive 21-Page Application Catalog

The Gramin Career Setu platform comprises 21 fully functional, accessible, and offline-cached pages:

| Route / Filename | Page Title | Functional Description & Technical Capabilities |
|---|---|---|
| **index.html** | Portal Gateway | Hero banner, key career pillars, bilingual switcher, offline cache status indicator. |
| **about.html** | About Project | Mission statement, student team details, Community Service Project objectives. |
| **careers.html** | Career Explorer | Multi-sector career catalog with filterable domain tags and average pay metrics. |
| **career-detail.html** | Career Deep-Dive | Detailed guides for specific professions, entrance exams, and required qualifications. |
| **jobs.html** | Job Opportunities | Searchable job listings, category filters, responsive pagination, direct apply buttons. |
| **job-detail.html** | Job Specification | Comprehensive role overview, eligibility criteria, hiring company profile, and application. |
| **quiz.html** | Aptitude Quiz | 12-question RIASEC psychometric quiz with bilingual text-to-speech audio read-aloud. |
| **quiz-results.html** | Aptitude Analysis | Interactive RIASEC radar chart, dimensional score breakdown, matched career cards. |
| **resume.html** | Resume Builder | 7-step dynamic wizard, live preview sidebar, 200-character objective counter, auto-save. |
| **resume-preview.html** | Resume Preview | Full A4 printable sheet rendering, dynamic ATS compatibility scoring, instant PDF export. |
| **roadmap.html** | Career Roadmaps | Visual education timelines from 10th standard to senior professional positions. |
| **scholarships.html** | Scholarships Hub | Directory of National and State merit/need scholarships with application deadlines. |
| **schemes.html** | Govt Schemes | Catalog of AP & Telangana welfare and youth skill schemes with documentation steps. |
| **workshops.html** | Skill Workshops | Hands-on vocational training schedules, ITI workshops, and online registration. |
| **resources.html** | Study Resources | Curated educational PDFs, entrance exam syllabi, and downloadable practice papers. |
| **stories.html** | Success Stories | Real-life inspirational journeys of rural students securing professional employment. |
| **profile.html** | Student Portfolio | Candidate personal profile, saved jobs, completed quiz history, and account data. |
| **dashboard.html** | User Dashboard | Active job applications, registered workshops, personalized recommendations hub. |
| **results.html** | Impact Dashboard | Platform deployment KPIs: students counseled, resumes built, partner villages reached. |
| **login.html** | Authentication | Bilingual login, role switcher (Student / Volunteer / Admin), phone OTP demo flow. |
| **admin/index.html** | Admin Console | Volunteer management, survey telemetry, two-way Google Sheets sync, storage meter. |
| **404.html** | 404 Error Handler | Bilingual fallback page with search box and direct navigation recovery links. |

## 5. Quality Assurance, Security Hardening & Test Suite

The project has undergone rigorous automated testing, security validation, and performance tuning to ensure bulletproof execution under field conditions:

- **Automated Test Suite (tests/project.test.js):** Executed via Node's native test runner (node --test tests/*.test.js). Verifies: 1) Entry-point delivery (HTTP 200); 2) Custom 404 handler for missing routes; 3) Encoded path traversal rejection (HTTP 403); 4) Malformed URI tolerance (HTTP 400 without crashing); 5) Jobs query parameter, category filtering, and pagination logic; 6) Resume objective 200-character boundary enforcement in both markup and JS engine.
- **Security Hardening (server.js):** Protected against path traversal attacks (/%2e%2e/, /..%2f) via strict normalized root jail enforcement. Malformed URI strings (%E0%A4%A) are handled safely with try/catch blocks without crashing the event loop.
- **Performance & Asset Optimization:** Media payload compressed by 84% (from 11.7 MB to 1.9 MB). Google Fonts replaced by self-hosted, subsetted WOFF2 fonts for Inter and Noto Sans Telugu (395 KB total), eliminating render-blocking external DNS queries.
- **Accessibility Compliance:** Engineered to WCAG 2.1 AA contrast standards. Features visible 2px focus rings, screen reader landmarks (main, nav, aside), and responsive touch targets (minimum 44x44px) for mobile usability.

## 6. Community Service Project Impact & Field Methodology

**Grassroots Implementation Workflow:** Gramin Career Setu was piloted in rural mandals as part of an active Community Service Project. Student volunteers organized weekend guidance camps at Zilla Parishad High Schools and Gram Panchayat offices. Using portable offline USB drives and local Wi-Fi hotspots, volunteers assisted 10th and 12th graders in identifying aptitude profiles, formatting ATS-compliant resumes, and applying for state welfare scholarships.

| Metric / KPI | Pilot Milestone | Observed Community Impact |
|---|---|---|
| **Students Guided** | 1,200+ Rural Youth | Completed RIASEC aptitude assessment and received tailored counseling. |
| **Resumes Compiled** | 450+ Formatted Resumes | Generated professional ATS-friendly PDF resumes for local jobs. |
| **Workshops Conducted** | 24 Field Sessions | Hands-on digital literacy and competitive examination workshops. |
| **Partner Villages** | 18 Gram Panchayats | Direct coordination with village sarpanches and youth clubs. |

## 7. Strategic Roadmap for Scalable Deployment

| Phase | Milestone | Key Technical Deliverables |
|---|---|---|
| **Phase 1 (Complete)** | Bilingual Core & PWA Engine | Full 21-page suite, Telugu translation, RIASEC quiz, resume builder. |
| **Phase 2 (Current)** | Field Testing & Pilot Camps | Pilot counseling drives in 5 mandals; gathering student usability feedback. |
| **Phase 3 (Next)** | Cloud API & Database Sync | PostgreSQL / Supabase backend, real-time employer postings, SMS OTP. |
| **Phase 4 (Future)** | Vernacular Voice AI Guidance | Offline AI audio counseling in colloquial Telugu dialects for low-literacy youth. |

**Conclusion:** Gramin Career Setu exemplifies the transformative power of purposeful technology in community service. By fusing modern web standards—offline PWA caching, atomic bilingual localization, psychometric aptitude modeling, and client-side document compilation—into an ultra-fast, zero-cost digital portal, the project empowers thousands of rural youth to overcome structural barriers and pursue their professional aspirations with dignity, clarity, and confidence.